

HCLTech-Style Online Assessment – Mock Test 1

Software Developer / Graduate Engineer Trainee Preparation

Total Questions	75
Total Time	90 Minutes
Difficulty	Moderate to Hard
Negative Marking	Practice simulation: No negative marking
Sections	Quantitative • Logical • Verbal • Technical • Pseudocode

Important: This is an HCLTech-style preparation mock, not an official HCLTech question paper. HCLTech assessment structure can vary by hiring program, role, and drive. The mock combines topics reported in recent HCLTech assessments with the current official AMP assessment areas.

ANSWER KEY

1. B 2. C 3. C 4. B 5. A 6. B 7. B 8. C 9. B 10. A 11. B 12. C 13. A 14. C 15. B
16. B 17. A 18. B 19. A 20. D 21. A 22. A 23. C 24. B 25. C 26. D 27. C 28. B 29. B 30. B
31. B 32. B 33. A 34. B 35. C 36. C 37. B 38. B 39. B 40. B 41. A 42. C 43. B 44. B 45. B
46. C 47. C 48. B 49. B 50. C 51. B 52. C 53. C 54. D 55. A 56. D 57. B 58. C 59. C 60. C
61. C 62. B 63. C 64. A 65. C 66. C 67. C 68. B 69. D 70. C 71. B 72. A 73. C 74. C 75. B

Trainer Note: Use 90 minutes as the target time. For a stronger simulation, enforce sectional timing and do not allow students to revisit a section after its time expires. The questions intentionally include calculation-heavy aptitude, multi-step reasoning, grammar traps, implementation-oriented technical MCQs, and code-tracing problems rather than basic recall.

SECTION A – QUANTITATIVE APTITUDE

1. A sum of money amounts to ■13,310 in 3 years at 10% compound interest per annum. What was the principal?
A) ■9,500
B) ■10,000
C) ■10,500
D) ■11,000
2. A and B can complete a task in 18 and 24 days respectively. They work together for 6 days, after which C completes the remaining work in 4 days. In how many days can C alone complete the work?
A) 8
B) 9
C) 10
D) 12
3. A train 180 m long crosses a platform in 24 seconds and crosses a man standing on the platform in 12 seconds. What is the platform length?
A) 150 m
B) 160 m
C) 180 m
D) 200 m
4. The ratio of incomes of A and B is 5:7 and their expenditures are in the ratio 3:5. If both save ■4,000, what is B's income?
A) ■12,000
B) ■14,000
C) ■16,000
D) ■18,000
5. A bag contains 5 red, 4 blue and 3 green balls. Two balls are drawn without replacement. What is the probability that they are of different colours?
A) 47/66
B) 49/66
C) 51/66
D) 53/66
6. If $x + 1/x = 5$, what is $x^3 + 1/x^3$?
A) 110
B) 115
C) 120
D) 125
7. The HCF of two numbers is 12 and their LCM is 720. If one number is 144, what is the other?
A) 48
B) 60
C) 72
D) 84
8. A mixture contains milk and water in the ratio 7:3. If 20 litres of water are added, the ratio becomes 7:5. What was the original quantity of the mixture?
A) 80 L
B) 90 L
C) 100 L
D) 120 L
9. A shopkeeper marks an article 40% above cost price and offers two successive discounts of 10% and 5%. What is the profit percentage?
A) 18.5%

- B) 19.7%
- C) 20%
- D) 21.5%

10. The average of 8 numbers is 42. If one number is replaced by 54, the average increases by 3. What was the original number?

- A) 24
- B) 27
- C) 30
- D) 32

11. How many 5-digit numbers can be formed using 0,1,2,3,4 without repetition and divisible by 5?

- A) 18
- B) 24
- C) 30
- D) 36

12. If $\log_{10}(x) + \log_{10}(x-2) = 3$, what is x?

- A) 2
- B) 3
- C) 4
- D) 6

13. A man travels 40% of a distance at 30 km/h and the remaining distance at 60 km/h. What is his average speed?

- A) 40 km/h
- B) 42 km/h
- C) 45 km/h
- D) 48 km/h

14. If 3 men or 5 women can complete a work in 24 days, how many days will 6 men and 10 women take?

- A) 4
- B) 5
- C) 6
- D) 8

15. The remainder when 7^{103} is divided by 6 is:

- A) 0
- B) 1
- C) 3
- D) 5

SECTION B – LOGICAL & ANALYTICAL REASONING

16. Find the next term: 3, 8, 18, 38, 78, ?

- A) 156
- B) 158
- C) 160
- D) 162

17. If COMPUTER is coded as DPNQVUFS, how is NETWORK coded?

- A) OFUXPSL
- B) OFUXPSK
- C) NFUXPSL
- D) OFUWPSL

18. Statements: All servers are machines. Some machines are virtual. No virtual machine is physical. Which conclusion definitely follows?

- A) Some servers are virtual
- B) Some machines are not physical
- C) No server is physical
- D) All machines are servers

19. A person walks 8 km north, 6 km east, 8 km south and 2 km west. How far is he from the starting point?

- A) 2 km
- B) 4 km
- C) 6 km
- D) 8 km

20. Five people P, Q, R, S and T sit in a row. P is left of Q but right of R. S is right of Q. T is between Q and S. Who sits in the middle?

- A) P
- B) Q
- C) R
- D) T

21. If 'A + B' means A is the mother of B, 'A – B' means A is the brother of B, and 'A × B' means A is the father of B, which expression means P is the maternal uncle of Q?

- A) $P - R + Q$
- B) $P + R - Q$
- C) $P \times R - Q$
- D) $P - R \times Q$

22. A statement is followed by two assumptions. Statement: 'Use our cloud service to reduce infrastructure costs.'

Assumptions: I. Infrastructure cost can be reduced using cloud services. II. The customer currently uses no cloud services.

Which is valid?

- A) Only I
- B) Only II
- C) Both I and II
- D) Neither

23. In a certain code, 246 means 'data is secure', 615 means 'secure systems work', and 583 means 'systems are scalable'.

Which digit represents 'secure'?

- A) 2
- B) 5
- C) 6
- D) 8

24. Six tasks A, B, C, D, E and F are scheduled. A must occur before C, B before D, and C before E. Which order is possible?

- A) C A B D E F
- B) A C E B D F
- C) B D C A E F
- D) A B E C D F

25. Data Sufficiency: Is n divisible by 12? I. n is divisible by 3. II. n is divisible by 4.

- A) I alone sufficient
- B) II alone sufficient
- C) Both together sufficient
- D) Either alone sufficient

26. Find the odd pair: 12:144, 15:225, 18:324, 21:441, 24:566

- A) 12:144
- B) 18:324
- C) 21:441
- D) 24:566

27. If all programmers are logical and some logical people are analysts, which statement is necessarily true?

- A) All analysts are programmers
- B) Some programmers are analysts
- C) All programmers are logical
- D) No analyst is logical

28. A clock gains 5 minutes every hour. If it is set correctly at 8:00 AM, what time will it show at 8:00 PM the same day?

- A) 8:45 PM
- B) 9:00 PM
- C) 9:05 PM
- D) 8:55 PM

29. Complete the analogy: Compiler : Source Code :: Database Engine : ?

- A) Network
- B) Query
- C) Operating System
- D) Hardware

30. In a queue, Rahul is 14th from the front and 19th from the rear. How many people are in the queue?

- A) 32
- B) 33
- C) 34
- D) 35

SECTION C – VERBAL ABILITY

31. Choose the grammatically correct sentence.
- A) Neither the manager nor the employees was aware of the change.
 - B) Neither the manager nor the employees were aware of the change.
 - C) Neither the manager or the employees were aware of the change.
 - D) Neither manager nor employees was aware of the change.
32. Choose the closest meaning of 'pragmatic'.
- A) Idealistic
 - B) Practical
 - C) Emotional
 - D) Uncertain
33. Identify the best replacement: 'The report comprises of five sections.'
- A) comprises five sections
 - B) is comprised five sections
 - C) comprise of five sections
 - D) has comprises five sections
34. Fill in the blank: The engineer insisted ____ reviewing the logs before deployment.
- A) at
 - B) on
 - C) for
 - D) to
35. Choose the antonym of 'mitigate'.
- A) Reduce
 - B) Relieve
 - C) Aggravate
 - D) Prevent
36. Choose the correctly spelled word.
- A) Accomodation
 - B) Acommodation
 - C) Accommodation
 - D) Accommadation
37. Choose the most appropriate sentence.
- A) Each of the developers have submitted their code.
 - B) Each of the developers has submitted their code.
 - C) Each developers has submitted their code.
 - D) Each of developers have submitted his code.
38. Identify the meaning of the idiom 'cut corners'.
- A) To work efficiently
 - B) To avoid unnecessary expenses by compromising quality
 - C) To complete a task early
 - D) To solve a difficult problem
39. Fill in the blank: Had the team tested the patch earlier, the outage ____ avoided.
- A) will have been
 - B) would have been
 - C) would be
 - D) had been
40. Choose the sentence with correct parallelism.

- A) The role requires coding, debugging, and to communicate with clients.
 - B) The role requires coding, debugging, and communication with clients.
 - C) The role requires to code, debugging, and communication.
 - D) The role requires coding, to debug, and communicating.
41. Select the best word: The evidence was ____ to support the hypothesis.
- A) insufficient
 - B) redundant
 - C) obsolete
 - D) ambiguous
42. Which sentence uses 'affect' correctly?
- A) The outage had a serious affect on users.
 - B) The outage effected the users badly.
 - C) The outage affected the users significantly.
 - D) The outage has an affect users.
43. RC: A distributed system may continue operating even when one component fails because responsibility can be replicated across multiple nodes. However, replication introduces consistency challenges because different nodes may temporarily hold different states. What is the central trade-off described?
- A) Speed versus memory
 - B) Availability/resilience versus consistency complexity
 - C) Security versus usability
 - D) Hardware versus software
44. RC: The passage states that replication can help a system survive component failure. Which inference is best supported?
- A) Replication eliminates all failures.
 - B) Replication can improve fault tolerance.
 - C) Replication guarantees identical states at all times.
 - D) Replication removes the need for monitoring.
45. Choose the best completion: 'Although the algorithm is asymptotically efficient, its practical performance may suffer ____ cache-unfriendly memory access.'
- A) because
 - B) due to
 - C) despite
 - D) unless

SECTION D – TECHNICAL FUNDAMENTALS

46. Which data structure is most appropriate for implementing an LRU cache with $O(1)$ average lookup and update?
- A) Stack only
 - B) Queue only
 - C) Hash map + doubly linked list
 - D) Binary search tree only
47. What is the average-case time complexity of searching for a key in a well-designed hash table?
- A) $O(\log n)$
 - B) $O(n)$
 - C) $O(1)$
 - D) $O(n \log n)$
48. Which SQL query correctly returns departments having more than 5 employees?
- A) `SELECT dept FROM emp WHERE COUNT(*) > 5 GROUP BY dept;`
 - B) `SELECT dept FROM emp GROUP BY dept HAVING COUNT(*) > 5;`
 - C) `SELECT dept FROM emp HAVING COUNT(*) > 5;`
 - D) `SELECT dept, COUNT(*) FROM emp WHERE COUNT(*) > 5;`
49. Which normal form primarily removes partial dependency on part of a composite key?
- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
50. Which scheduling algorithm may cause starvation if priorities are not adjusted?
- A) FCFS
 - B) Round Robin
 - C) Priority Scheduling
 - D) FIFO
51. Which OSI layer is primarily responsible for end-to-end reliable delivery?
- A) Network
 - B) Transport
 - C) Session
 - D) Data Link
52. In OOP, method overriding primarily demonstrates:
- A) Encapsulation
 - B) Compile-time polymorphism
 - C) Runtime polymorphism
 - D) Abstraction only
53. Which statement about an interface in Java is correct?
- A) A class can extend multiple classes.
 - B) A class can implement multiple interfaces.
 - C) An interface cannot contain methods.
 - D) Interfaces must always be instantiated directly.
54. What is the worst-case time complexity of quicksort with a consistently poor pivot choice?
- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(n \log n)$
 - D) $O(n^2)$
55. Which SQL operation is used to combine rows from two tables based on a related condition?

- A) JOIN
- B) GROUP BY
- C) UNION ALL only
- D) ORDER BY

56. Which of the following can lead to a deadlock?

- A) Mutual exclusion
- B) Circular wait
- C) Hold and wait
- D) All of the above

57. A binary search tree is considered balanced when its height is approximately:

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

58. Which protocol is connection-oriented and provides reliable, ordered byte-stream delivery?

- A) UDP
- B) IP
- C) TCP
- D) ARP

59. Which key uniquely identifies a row and cannot contain NULL values in a relational table?

- A) Foreign key
- B) Candidate attribute
- C) Primary key
- D) Secondary key

60. Which memory is generally closest to the CPU and fastest among the following?

- A) HDD
- B) RAM
- C) Cache
- D) Optical disk

SECTION E – PSEUDOCODE / PROGRAMMING LOGIC

61. What is printed? `x=1; for i=1 to 4: x=x*i; print(x)`
- A) 4
 - B) 12
 - C) 24
 - D) 120
62. What is the output? `a=5; b=2; a=a+b; b=a-b; a=a-b; print(a,b)`
- A) 5 2
 - B) 2 5
 - C) 7 5
 - D) 3 2
63. What is the final value of count? `count=0; for i=1 to 5: for j=i to 5: count=count+1`
- A) 10
 - B) 12
 - C) 15
 - D) 20
64. What does this algorithm return for [2, 7, 11, 15] and target 9? `for i=0..n-1: for j=i+1..n-1: if a[i]+a[j]==target return (i,j)`
- A) (0,1)
 - B) (0,2)
 - C) (1,2)
 - D) (-1,-1)
65. What is the output? `s='HCLTECH'; print(s[1:5])`
- A) HCLT
 - B) CLTE
 - C) CLT
 - D) LTEC
66. A loop repeatedly sets $n = n/2$ until n becomes 1. For $n=1024$, how many divisions occur?
- A) 8
 - B) 9
 - C) 10
 - D) 11
67. What is the time complexity of: `for i=1 to n: j=1; while j<n: j=j*2`
- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(n \log n)$
 - D) $O(n^2)$
68. What is printed? `a=[1,2,3,4,5]; reverse(a,1,3); print(a)` Assume reverse reverses indices 1 through 3 inclusive.
- A) [1,2,3,4,5]
 - B) [1,4,3,2,5]
 - C) [1,4,3,2,5]
 - D) [1,5,4,3,2]
69. A recursive function returns $f(n)=n*f(n-1)$, with $f(1)=1$. What does $f(5)$ return?
- A) 15
 - B) 24
 - C) 100
 - D) 120

70. What is the output? `x=0; for i=1 to 5: if i%2==0: x=x+i; else: x=x-1; print(x)`
- A) 0
 - B) 1
 - C) 2
 - D) 3
71. Which technique is most appropriate to find the first non-repeating character in a string in $O(n)$ time?
- A) Nested loops only
 - B) Frequency map + second pass
 - C) Sorting only
 - D) Binary search
72. An array is [4, 1, 3, 2]. After one complete pass of standard bubble sort in ascending order, what is the array?
- A) [1,3,2,4]
 - B) [1,4,2,3]
 - C) [4,1,2,3]
 - D) [1,2,3,4]
73. What is the output? `x=10; y=3; print(x//y + x%y)`
- A) 3
 - B) 4
 - C) 5
 - D) 6
74. Which approach solves the classic Two Sum problem in $O(n)$ average time?
- A) Nested loops
 - B) Sorting with two pointers only
 - C) Hash map
 - D) Binary search without sorting
75. What is the key risk if an iterative binary search updates `low=mid` instead of `low=mid+1` when `a[mid]` is too small?
- A) Integer overflow always occurs
 - B) The algorithm may fail to make progress and loop indefinitely
 - C) The array becomes unsorted
 - D) Search becomes $O(1)$